

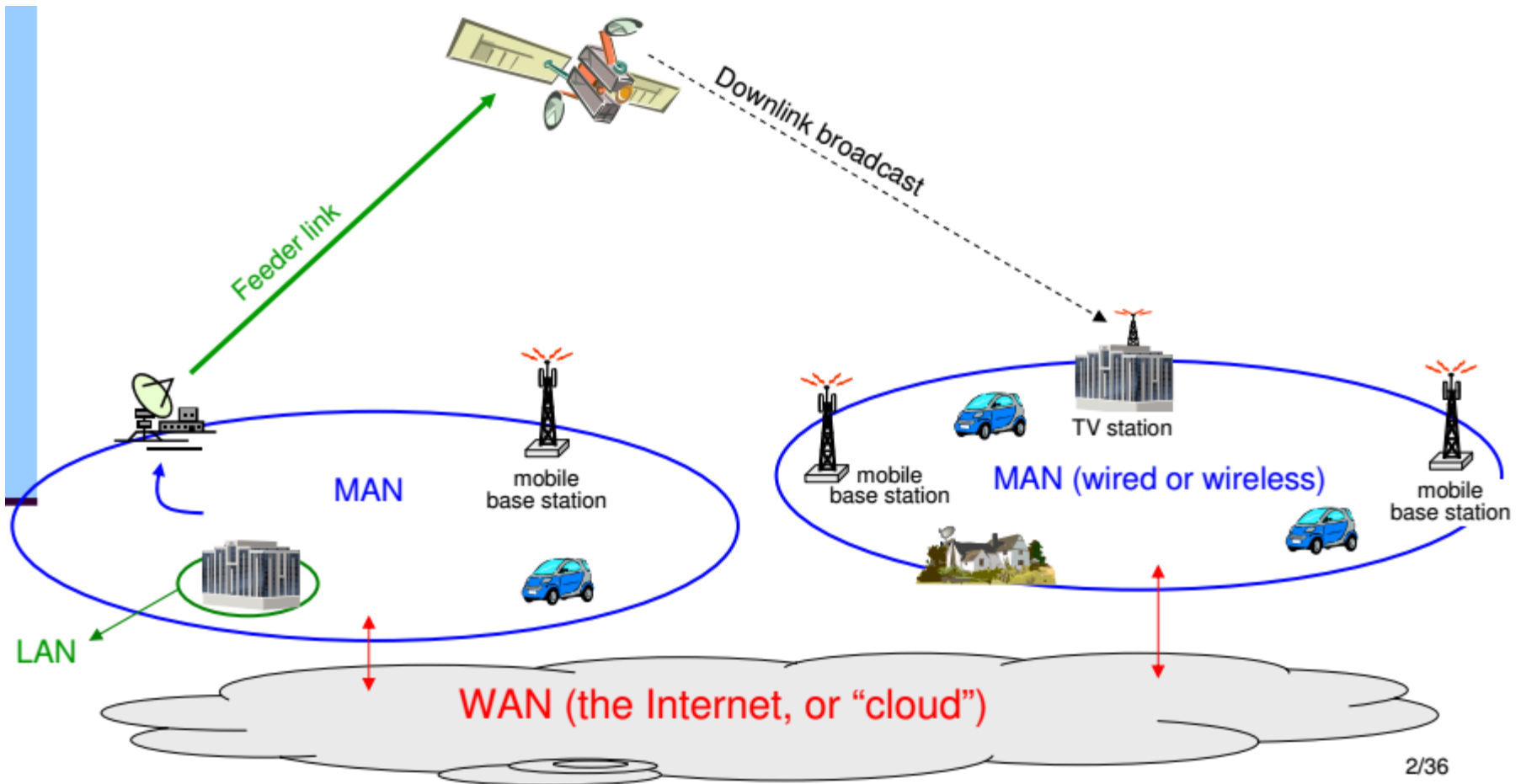
Computer Networks

Lectures 6 and 7

Networking and the Internet

- Network Fundamentals
- The Internet
- The World Wide Web
- Internet Protocols
- Security

A Networked World

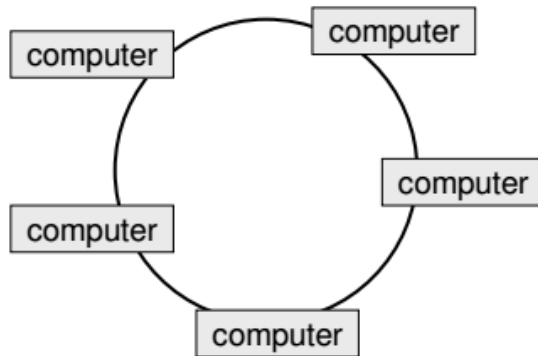


Network Classifications

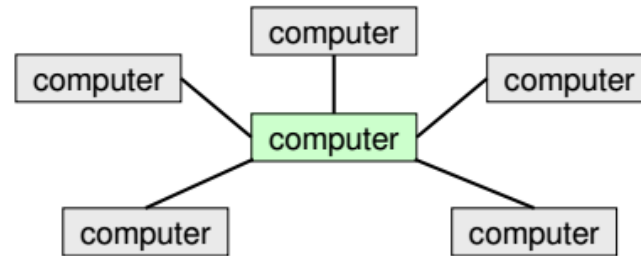
- Geographic Scope
 - Personal area network (PAN)
 - Local area network (LAN)
 - Metropolitan area (MAN)
 - Wide area network (WAN)
- Ownership
 - Closed versus open
- Topology (configuration)
 - Ring
 - Bus (Ethernet)
 - Star (Wireless networks with central Access Point)

Network topologies

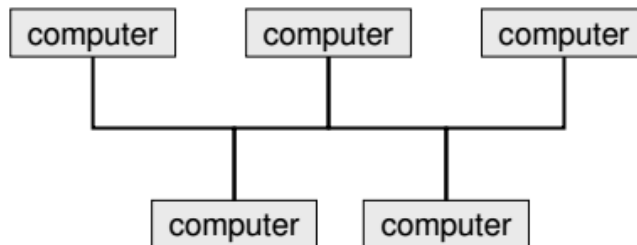
❑ Ring:



❑ Star:



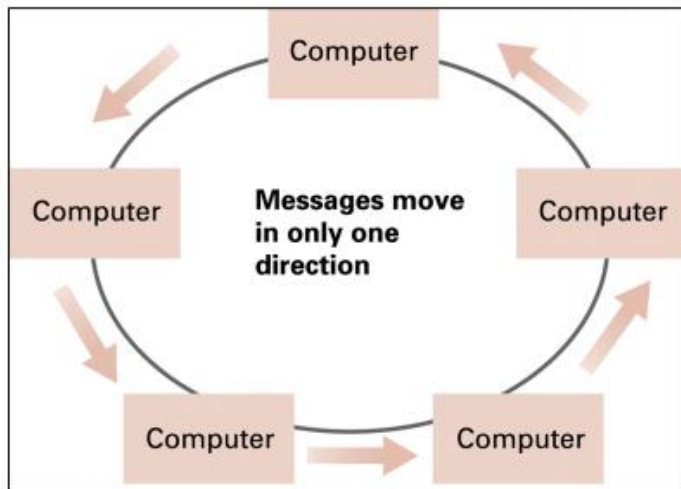
❑ Bus:



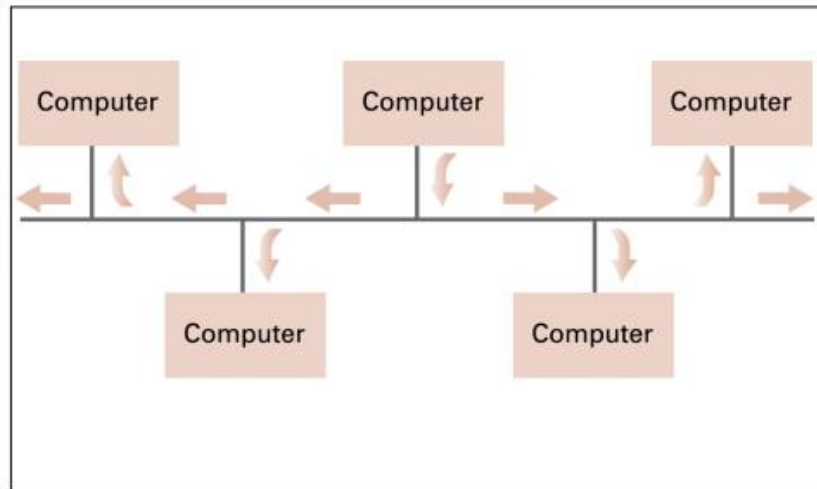
The star configuration is popular today since wireless networks use this configuration; the central computer is called access point (AP) or base station (BS)

Communication over Network

- The topology of a network determines its communication protocol
- A **network protocol** is a set of rules for computers to perform data exchange



Ring Network



Bus Network

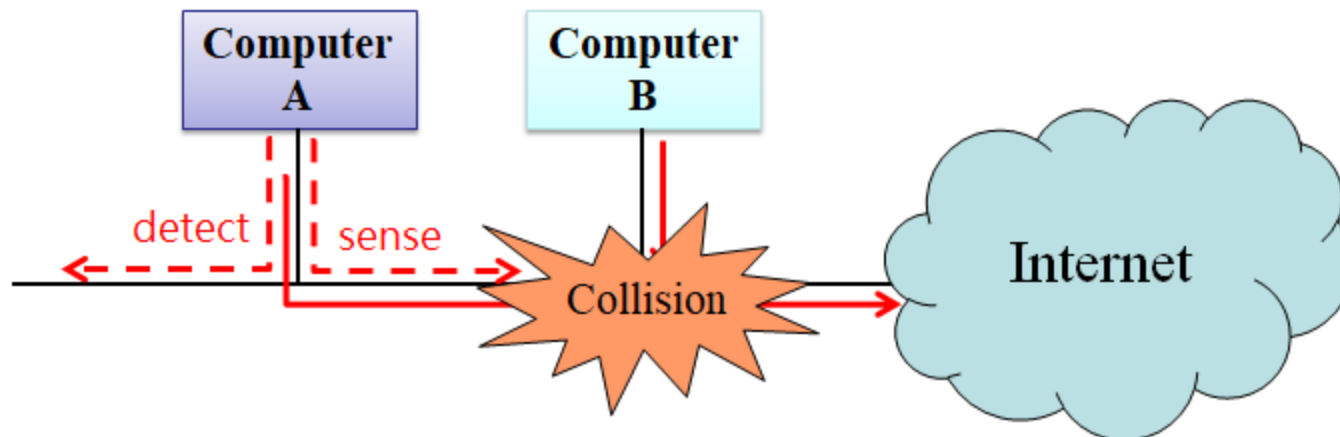
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Token Ring Protocol

- Developed by IBM in 1970
- The computer that receives the 'token' message can become the originating computer of a data message
- Data messages are transmitted in one direction
 - Each computer on the ring topology forwards the message to the next one
 - Only the destination machine keeps a copy of the data message before it forwards the message
 - The origination computer of the message will not forward it again when it received the message

Ethernet Protocol for Bus Network

- Ethernet protocol uses the rule called “Carrier Sense, Multiple Access with Collision Detection (CSMA/CD)”
- Each computer on the network has an address ID
- Any messages are broadcasted to all computers on the network
- A computer only keeps the message addressed to it
- If more than one computers try to send messages at the same time, they will all detect the “collision” and stop sending messages.
- They will try again later, after a random period of waiting.



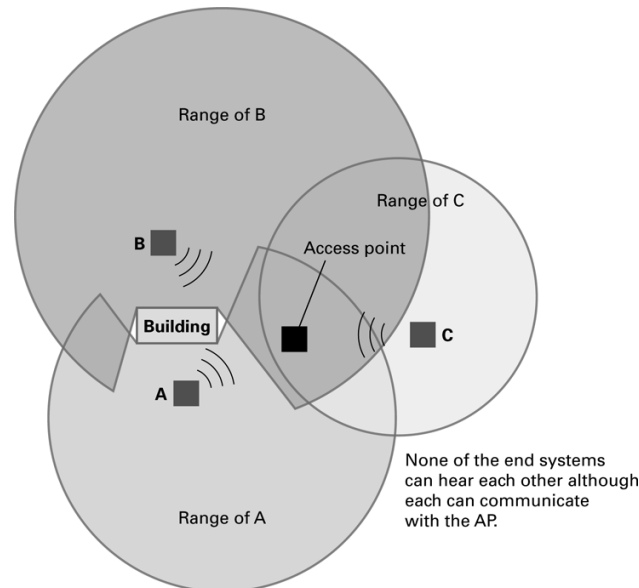
CSMA/CD Algorithm

1. Has any data to send? If yes,
 - 1.1 Check if medium to send the data is idle. If not, wait until it becomes ready.
 - 1.2 Start transmitting and monitor for collision during transmission.
 - 1.3 Did a collision occur? If yes,
 - 1.3.1 Continue transmission with **Jam Signal** until minimum packet time is reached to ensure all receivers detect collision
 - 1.3.2 Increment retransmission counter. If maximum number of transmission attempts reached, abort transmission.
 - 1.3.3 Wait for random amount of time
 - 1.3.4 Go to step 1.1
2. Reset retransmission counters and go to step 1

Note: Collision detection methods differ based on each medium. For electrical bus, it can be comparing transmitted and received data, or recognizing higher than normal amplitude on bus

Wireless Protocol

- Wireless network uses the rule called “Carrier Sense, Multiple Access with Collision Avoidance (CSMA/CA)” which is standardized by IEEE 802.11 standard, also referred as **WiFi**
- A “newcomer” must wait until it does not “hear” any messages before it can *request* to use the channel
- To avoid the “**hidden terminal problem**” the newcomer must wait for the AP to grant its request before it sends messages



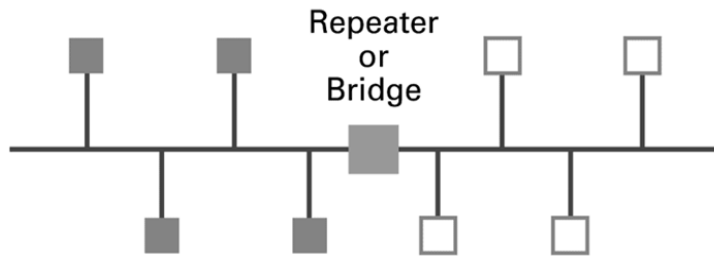
CSMA/CA Algorithm

1. Listen to the shared medium? If no signal heard,
 - 1.1 Transmit Request to Send (RTS) signal
 - 1.2 If Clear to Send (CTS) signal received from AP,
 - 1.2.1 Transmit data
 - Else (CTS not received):
 - 1.2.2 Wait for random amount of time
 - 1.2.3 Go to step 1
- Else (signal heard):
 - 1.3 Wait for random amount of time
 - 1.4 Go to step 1

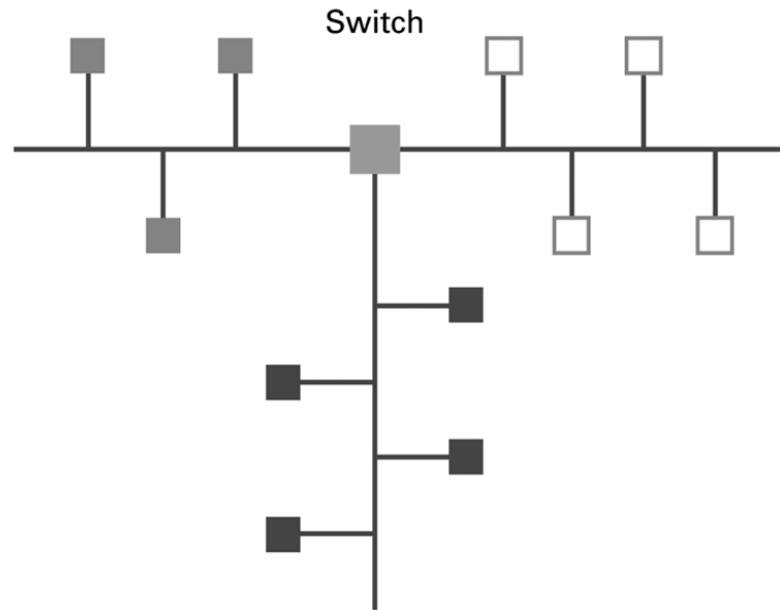
Connecting Networks

- **Repeater:** Extends a network, amplifies its signals and does not care about their meaning
- **Bridge:** Connects two compatible networks. It passes the message to other network, only if it is destined for some computer there, otherwise it blocks it.
- **Switch:** Connects several compatible networks. It is a bridge with multiple connections.
- **Router:** Connects two incompatible networks resulting in a network of networks called an **internet**. It is a special purpose computer used for forwarding messages.

Building a large bus network from smaller ones

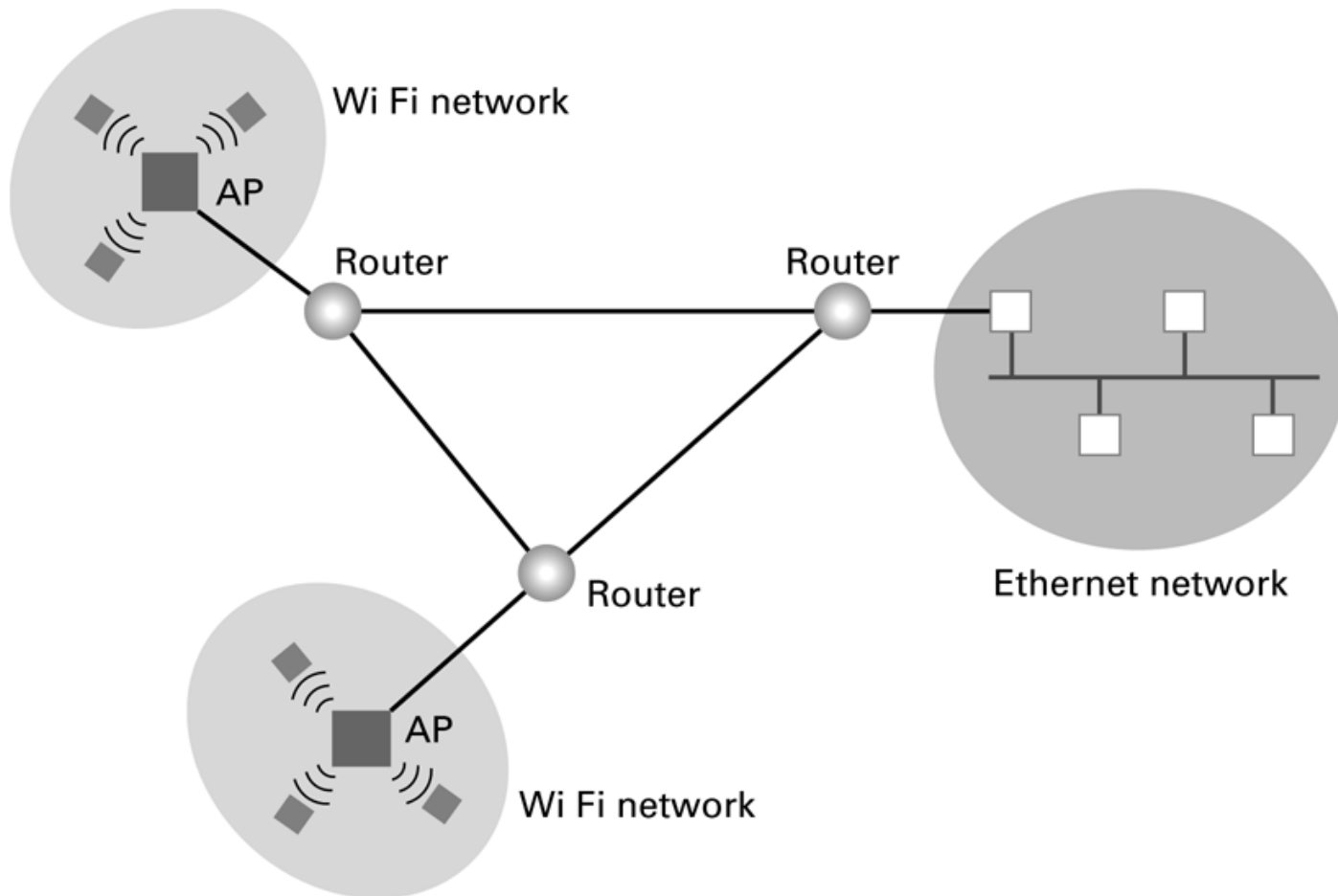


a. A repeater or bridge connecting two buses



b. A switch connecting multiple buses

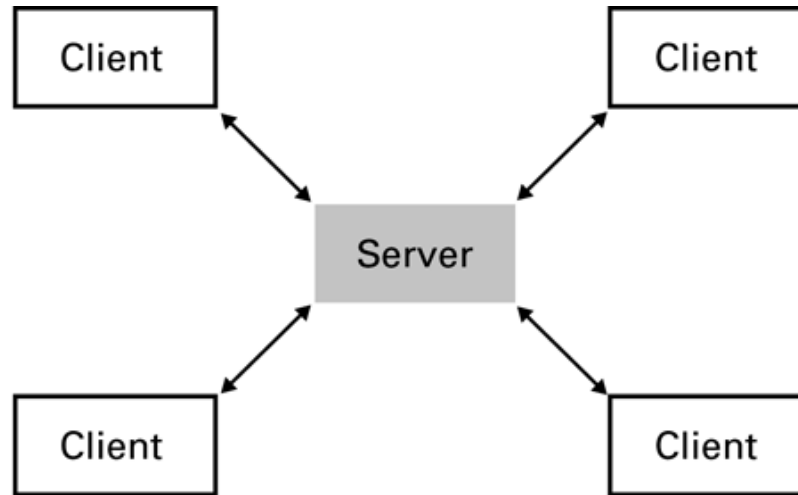
Routers connecting two WiFi networks and an Ethernet network to form an internet



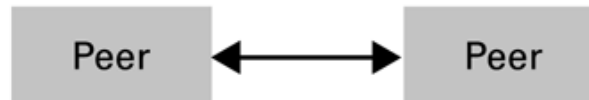
Inter-process Communication

- Client-server
 - One server, many clients
 - Server must execute continuously
 - Client initiates communication
 - Examples: print server, file server
- Peer-to-peer (P2P)
 - Two processes communicating as equals
 - Peer processes can be short-lived
 - Peer-to-peer **network** is a misuse of the terminology

The client/server model compared to the peer-to-peer model



a. Server must be prepared to serve multiple clients at any time.



b. Peers communicate as equals on a one-to-one basis.

Distributed Systems

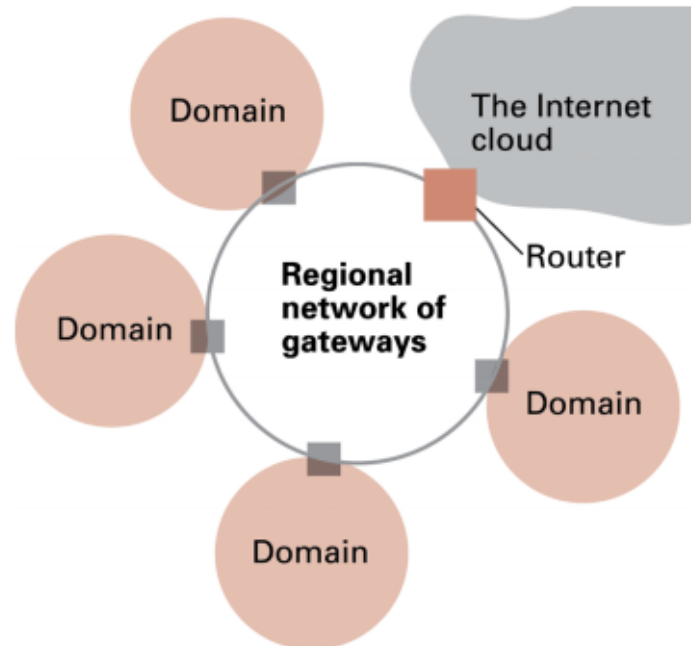
- A distributed system is a system with software units that run on different computers across the network.
- Each software unit is a process running on a host (i.e. a computer on the network)
- Distributed processes usually have to perform data exchange, remote invocation, synchronization, etc.
- Various distributed computing models:
 - **Cluster computing** – tightly-coupled, supercomputer style
 - **Grid computing** – loosely-coupled, SETI@HOME style
 - **Cloud computing** – pools of shared computers maintained by private companies
 - Amazon's Elastic Compute Cloud
 - Google Drive

The Internet

- The Internet: An internet that spans the world
 - Started by DARPA in 1970's
 - Original goal was to develop a means of connecting networks that would not be disrupted by local disasters
 - Today a commercial undertaking that links a worldwide combination of PANs, LANs, MANs, and WANs involving millions of computers
- The Internet is a ***packet-switched*** network. That is, all the data are transmitted on a packet-by-packet basis.
- Another type of network is called ***circuit-switched***, which means a virtual connection will be established before the transmission of data.

Internet Architecture

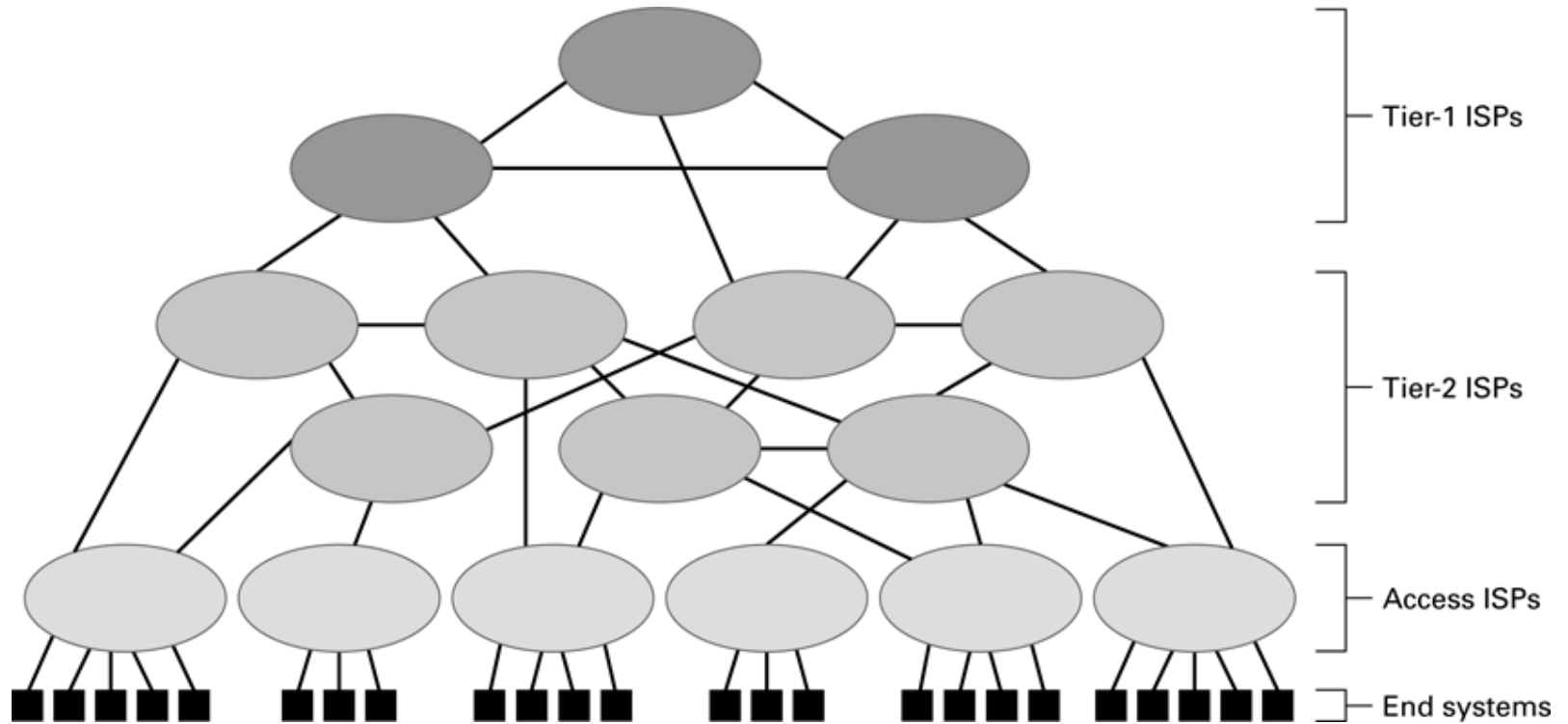
- ❑ The Internet is a collection of domains
 - Each domain is a network or a set of inter-connected networks controlled by a single organization
 - Domains must be registered through ICANN, Internet Corporation for Assigned Names & Numbers
- ❑ A gateway is a router connecting a domain to the rest of the Internet (referred to as the ***cloud***)



Internet Composition

- Internet Service Provider (ISP)
 - Tier-1: e.g. AT&T, CenturyLink, Telia Carrier, etc
 - Tier-2: e.g. UzbekTelecom, British Telecom
- Access or tier-3 ISP: Provides connectivity to the Internet,
 - Hot spot (wireless)
 - Telephone lines
 - Cable/Satellite systems DSL
 - Fiber optics
 - E.g. Sarkor Telecom, TPS, UzOnline, etc

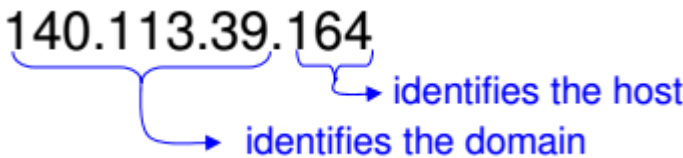
Internet Composition



Internet Addressing

- Each machine on a network must have a unique address: for the Internet, this is called the Internet Protocol (IP) address
 - For IPv4, an IP address is a 32-bit identifier for a machine
 - For IPv6, an IP address is 128-bits
- IP address is often written in dotted decimal notation:

– IPv4 example: 140.113.39.164



identifies the host
identifies the domain

– IPv6 example: fe80::3153:525f:6964:8d84

or

fe80:0000:0000:0000:3153:525f:6964:8d84

Internet Textual Addressing

- IP addresses are difficult for human to remember
- Each IP may have an equivalent mnemonic address, which is composed of a domain name and a host name (e.g. `eclass.inha.uz`)
- Domain name is the part assigned by a registrar
 - Top level domain (TLD) is the classification of domain owner (e.g. `.com`, `.uz`)
 - A domain name server (DNS) on the network translates the mnemonic addresses to binary IP addresses
- Host name is assigned by domain administrator
- Domain owner must run a DNS in order for other computers to find him/her
- Or else, he/she has to hire a DNS maintained by an ISP.

Internet Corporation for Assigned Names & Numbers (ICANN)

- ICANN is a nonprofit organization that is responsible for coordinating the maintenance and procedures of several databases related to the namespaces of the Internet.
- It allocates IP addresses to ISPs who then assign those addresses within their regions.
- Oversees the registration of domains and domain names by maintaining central Internet address pools and ***DNS Root*** registries.



Application Protocols

- Network News Transfer Protocol (NNTP) – protocol used for exchanging forum posts
- File Transfer Protocol (FTP) – protocol used for copying/moving/deleting files
- Telnet and SSH – protocol used for accessing remote computers
- Hypertext Transfer Protocol (HTTP) – protocol used for transporting web pages

Email Protocols

- ***Simple Mail Transfer Protocol*** (SMTP) defines a way that two computers on the network may interact when transmitting an email message from one host to the other.
 - Mail servers use SMTP to relay received emails to neighboring servers.
 - Although SMTP has redundant text sent during the communication, its status codes and commands are strictly specified
 - Some of the commands involve:
 - **HALO, MAIL, RCPT, DATA, QUIT**
 - Modern SMTP also uses additional commands to check the authenticity of sender host to fight spammers/phishers.

SMTP Simple Mail Transfer Protocol

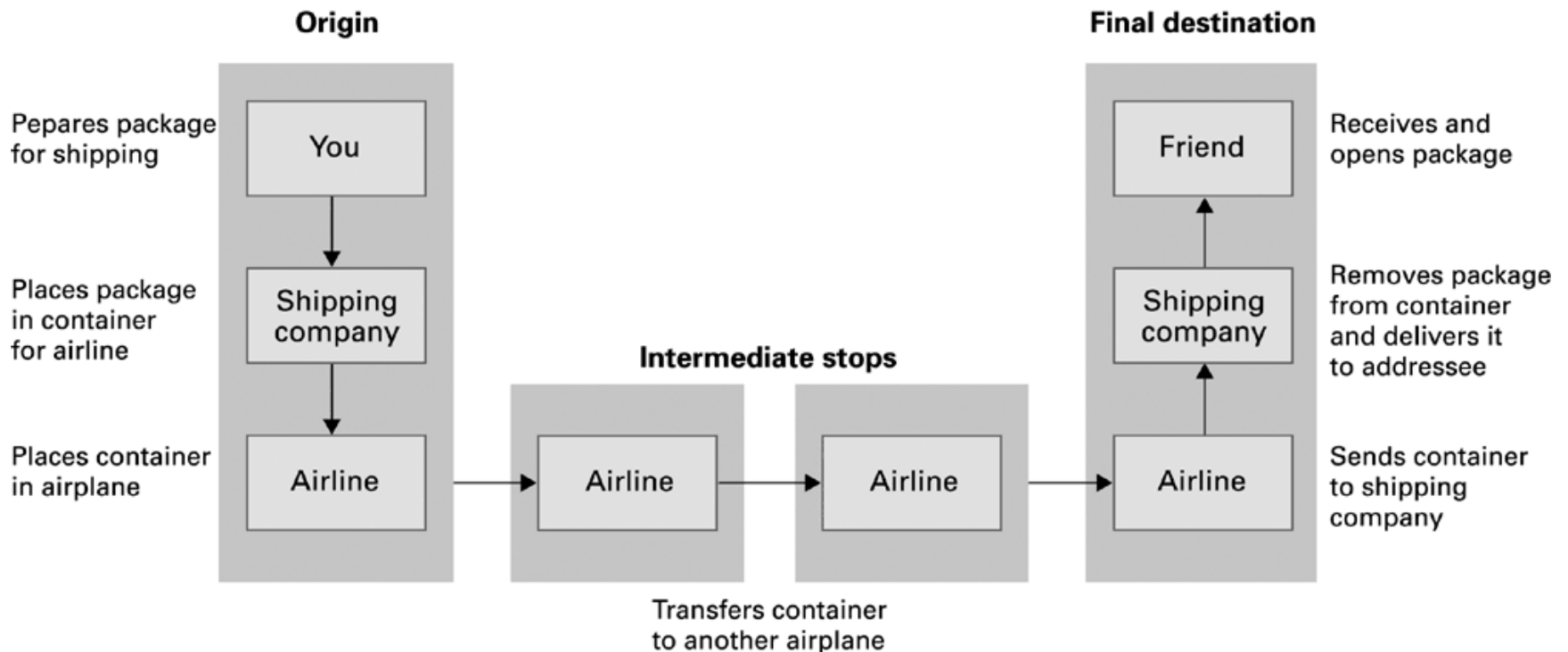
```
220 mail.tardis.edu SMTP Sendmail Gallifrey-1.0; Fri, 23 Aug 2413
14:34:10
HELO mail.skaro.gov
250 mail.tardis.edu Hello mail.skaro.gov, pleased to meet you
MAIL From: dalek@skaro.gov
250 2.1.0 dalek@skaro.gov... Sender ok
RCPT To: doctor@tardis.edu
250 2.1.5 doctor@tardis.edu... Recipient ok
DATA
354 Enter mail, end with "." on a line by itself
Subject: Extermination.
EXTERMINATE!
Regards, Dalek
.
250 2.0.0 r7NJYAE1028071 Message accepted for delivery
QUIT
221 2.0.0 mail.tardis.edu closing connection
```

More Email Protocols

- ***Multipurpose Internet Mail Extensions*** (MIME) converts non-ASCII data to SMTP compatible form, thus enables sending file attachments along with email body.
- ***Post Office Protocol v3*** (POP3) provides “download” access to client, so it can copy email files to its hard-disk, and then use email software to open it.
- ***Internet Mail Access Protocol*** (IMAP) provides browsing, reading, editing, removing permissions to its clients.
 - Client does not need to download the whole file in order to read the email.

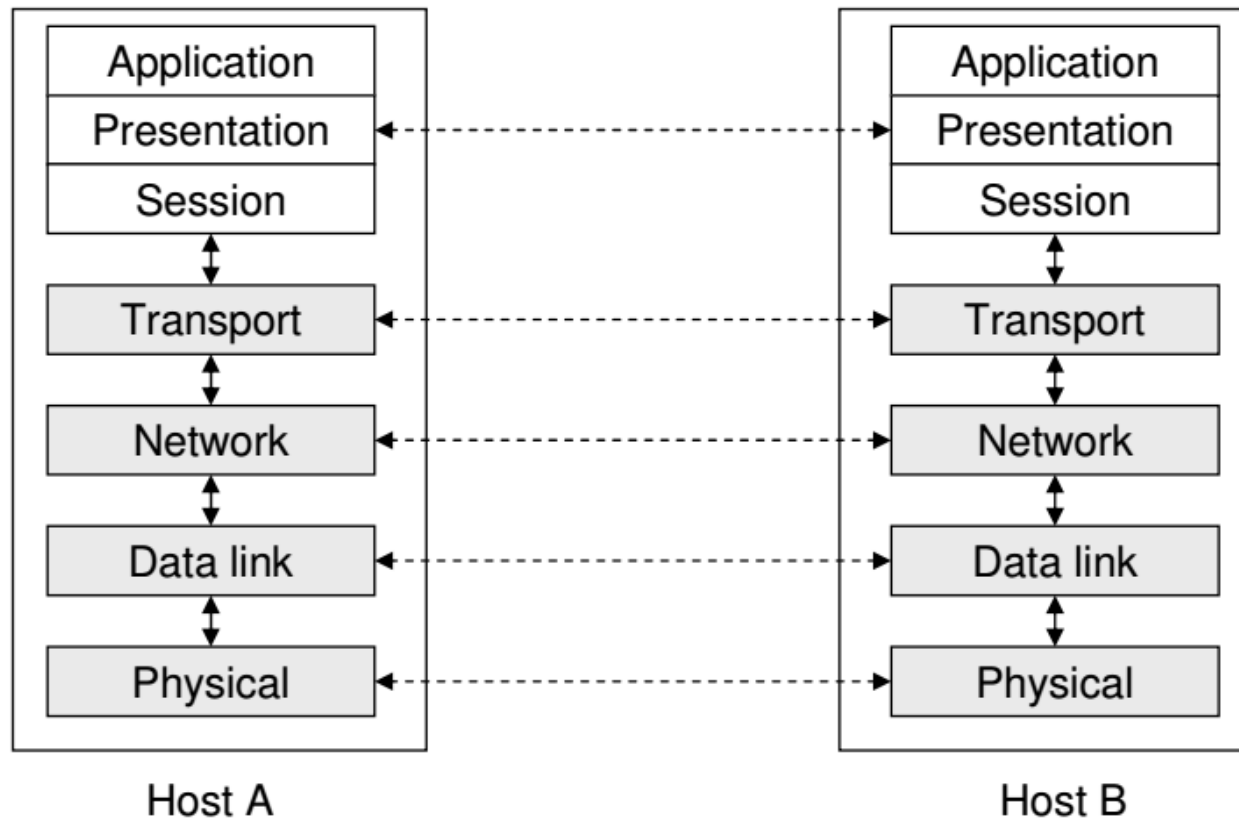
Internet Protocols

Package-shipping example



ISO's 7-layer model

- Most networks are designed using International Organization for Standardization (ISO)'s 7-layer model



Physical Layer

- Physical layer describes the electrical/optical/mechanical and functional interfaces to the physical medium, pass the signals for all higher layers
- Data encoding:
 - what represents 1 and what represents 0
 - At what interval bits are delimited?
- How data frame is delimited?

Data Link Layer

- The data link layer provides error-free transfer of data frames from one node to another over the physical layer, allowing layers above to assume virtually error-free transmission over the link
- Logical link establishment/termination between two nodes
- Frame traffic control prevents transmitting nodes to wait until already sent frame is processed by data link layer
- Delimiting the boundaries of frames
- Error checking to ensure the integrity of frame

Network Layer

- Network layer controls the operation of the subnet, deciding which physical path the data should take based on network conditions, priority of service and other factors
- It routes the traffic to the right network
- Fragmentation of data into data frames
- Translates logical address to physical address

Transport Layer

- Message segmentation and desegmentation
- Message acknowledgement provides reliable end-to-end message delivery with acknowledgements
- Message traffic control stops transmitting machine from sending the next message if not all the segments of the message were received

Session Layer

- The session layer allows session establishment between processes running on different stations.
- Session establishment, maintenance and termination allows two processes on different machines to establish, use and terminate connection called a session
- Session support performs the functions such as authentication and security

Presentation Layer

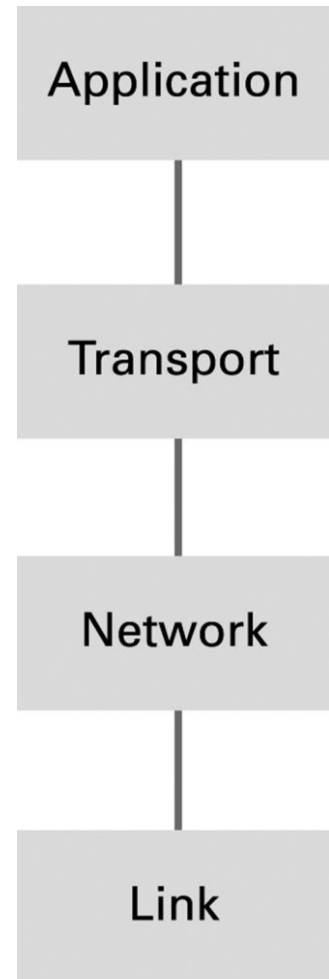
- Character code translation
- Data conversion: integer, floating point, so on
- Data compression: reduces the number of bits that need to be transmitted on the network
- Data encryption for security purposes

Application Layer

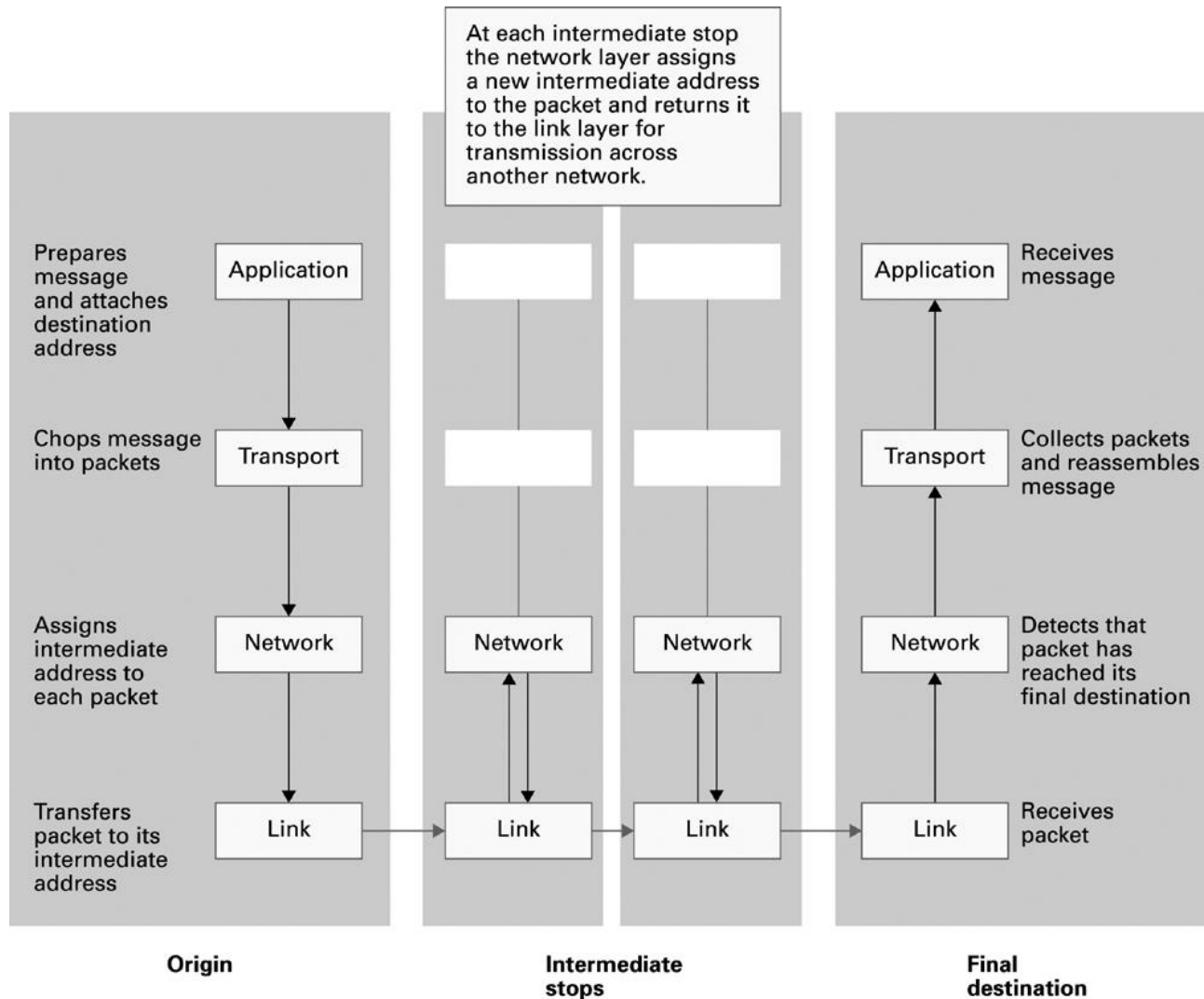
- Application layer serves as the window for users and application process to access network services
- This layer contains a variety of common application-level protocols such as:
 - Shared printers
 - Emails
 - File server
 - Chats

Internet Software Layers

- The Internet, just like the ISO's 7-layer model, also uses layered protocol approach.
- **Application:** Constructs message with address
- **Transport:** Chops message into packets
- **Network:** Handles routing through the Internet
- **Link:** Handles actual transmission of packets



Following a message through the Internet



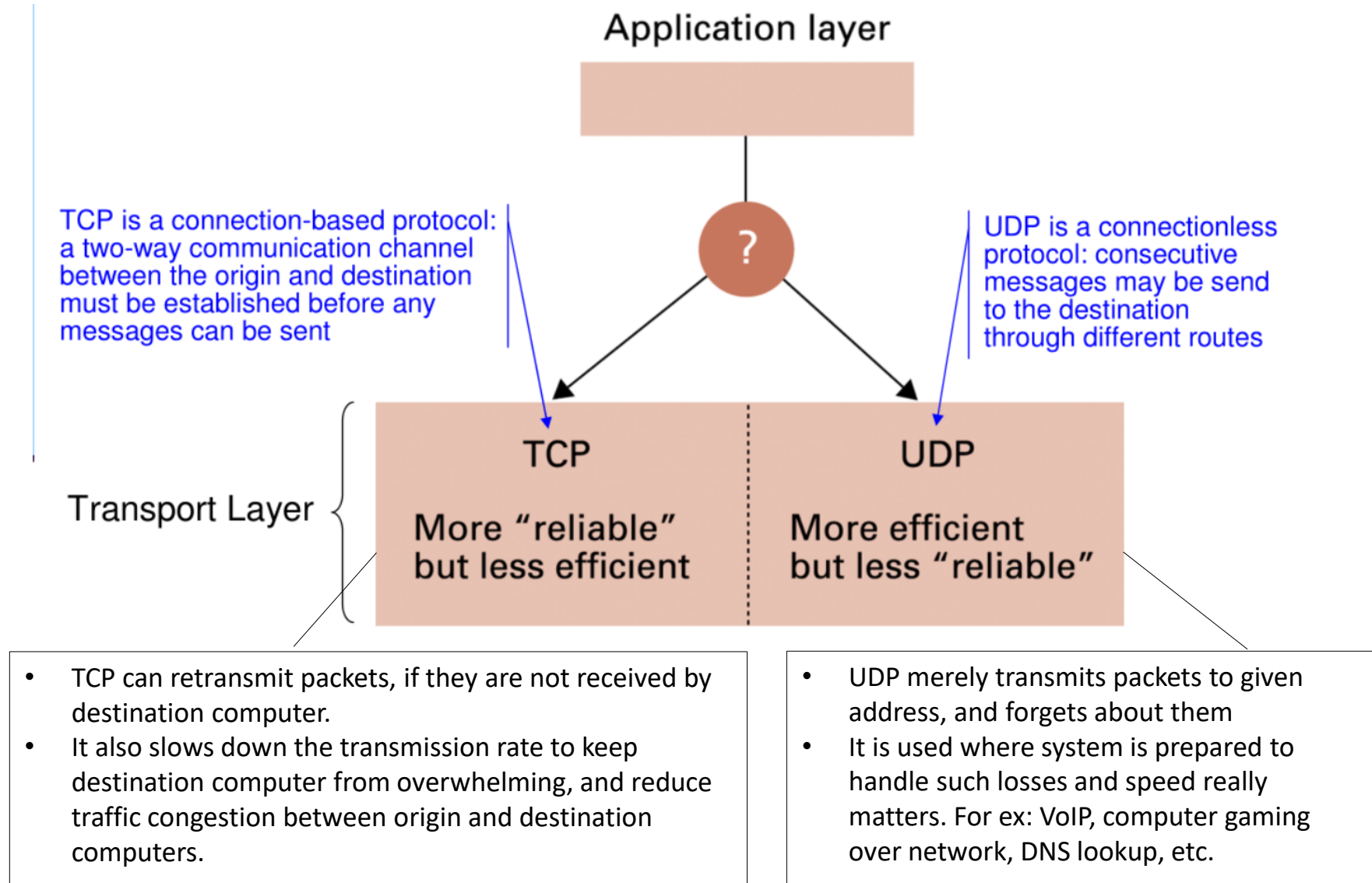
Port Number

- A computer have a unique IP address, all applications running in this computer share the same IP address
- When an Internet packet arrives at an IP address, which application on the computer should receive it?
 - Port numbers are used to identify application that should receive the reassembled message from Transport Layer.
- Web pages transferred over HTTP are commonly sent to port 80, so the web server listens to this port for HTTP requests coming from browsers.
- Browser, on the other hand, listens to an arbitrary port for responses coming from web server.
- SMTP server listen to port 25 to process commands related to mail transfer.

TCP/IP Protocol Suite

- Application Layer
 - HTTP, FTP, SMTP, POP3, IMAP, etc
- Transport Layer
 - Transmission Control Protocol (TCP)
 - User Datagram Protocol (UDP)
- Network Layer
 - Internet Protocol (IP)
 - IPv4
 - IPv6
- Data Link
 - Ethernet – Collision Detection Protocol
 - Wi-Fi – Collision Avoidance Protocol
 - Token Ring

Choosing between TCP and UDP



Internet Protocol

- IP is the Internet's standard for ***forwarding*** and ***routing*** the packets:
 - Forwarding means relaying packets through Internet
 - Routing means updating the layer's of forwarding table to reflect the changing conditions of the network.
- Above tasks are helpful because some of the routers in Internet might not be operational, or some parts of Internet can be congested, so IP protocol has to find alternative paths to transmit packages.
- IP appends special ***hop count (aka time to live, TTL)*** to each packet before sending it.
 - Hop count is initially set to 64.
 - Every time a packet reaches the next intermediate router, that router reduces the hop count by 1.
 - If hop count of the packet becomes 0, it is not forwarded anymore, thus preventing the possibility for a packet to travel the network eternally.

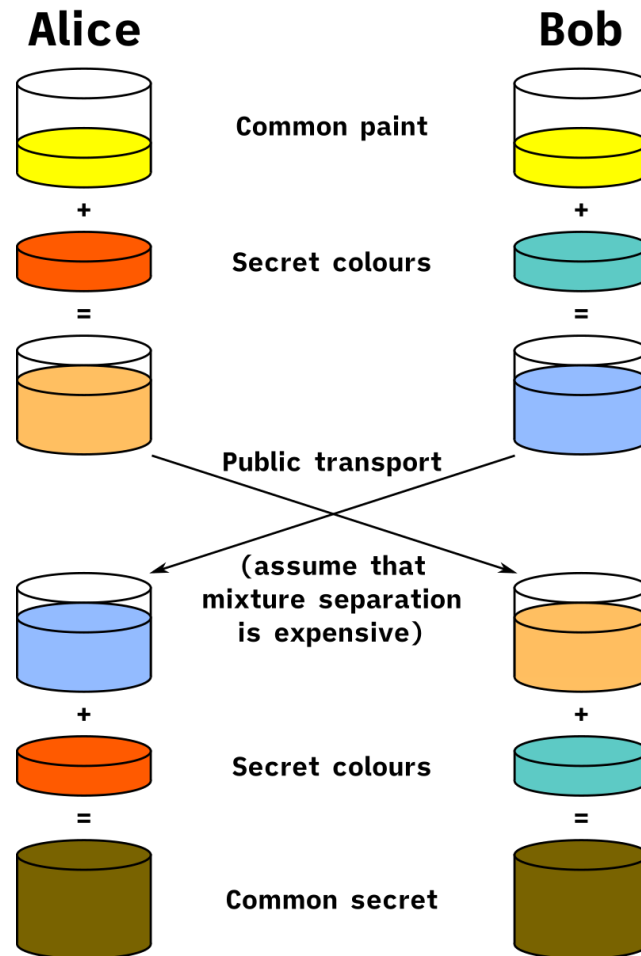
Security

- Attacks
 - Malware (viruses, worms, Trojan horses, spyware, phishing software)
 - Distributed Denial of Service (DDoS)
 - Spam
- Protection
 - Firewalls
 - Spam filters
 - Proxy Servers
 - Antivirus software

Privacy of Communication

- In old days, most data sent over the Internet were unprotected
 - Anyone with a network sniffer could read it
- Today, encryptions are used to protect your privacy
 - Secure versions of network protocols such as FTPS, HTTPS and SSL are used to transmit data with encryption
 - This is due to the emergence of ***asymmetric encryption*** methodologies available.
- In symmetric encryption, you can use one key to encrypt a message, and use the same key to decrypt it.
 - Knowing how the message has been encrypted is enough to decrypt it.
- In asymmetric encryption, you use one key to encrypt the message, but another key to decrypt it
 - Knowing how the message has been encrypted is not enough to decrypt it.

Diffie–Hellman Key Exchange Algorithm



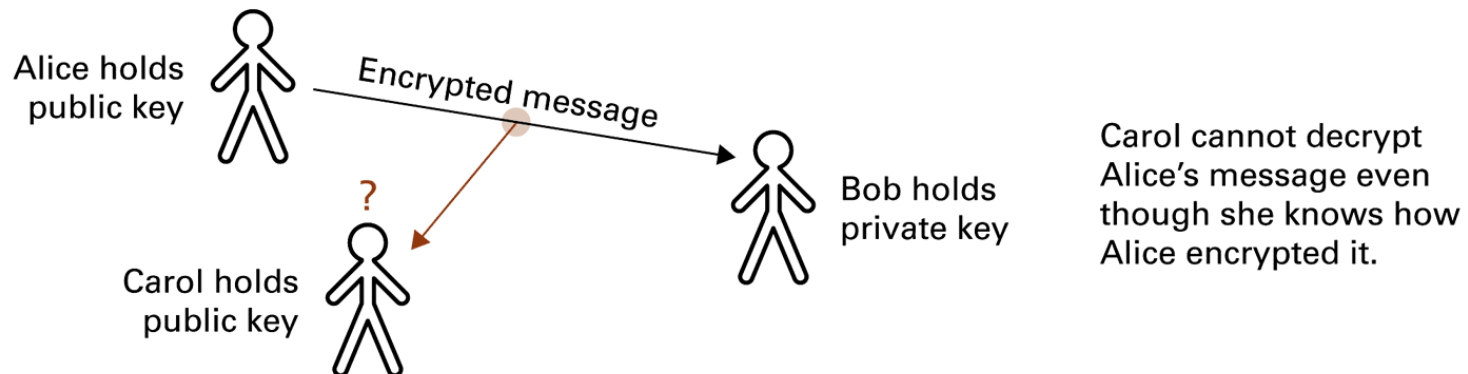
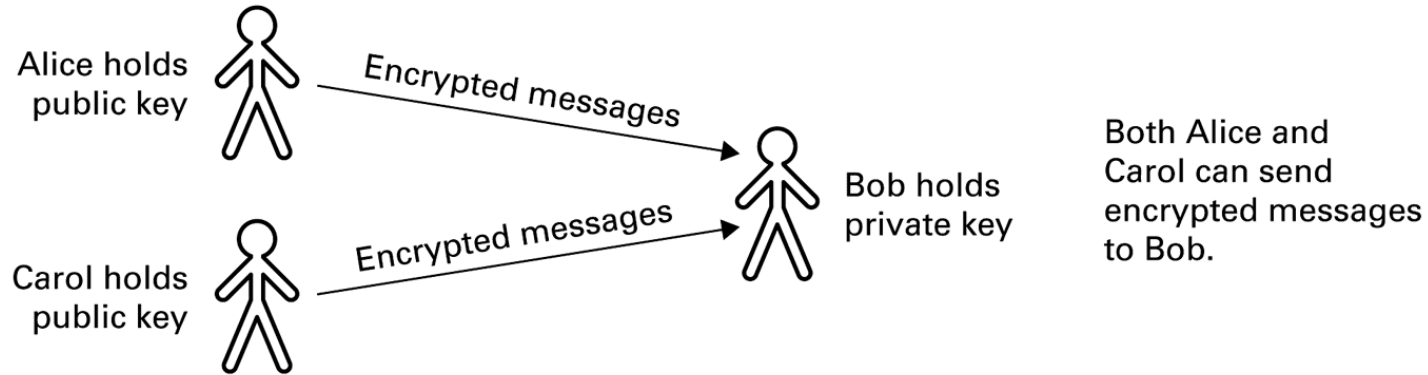
Cryptographic Explanation

1. Alice and Bob publicly agree to use $p = 23$ and $g = 5$.
2. Alice chooses a secret integer $a = 4$, then sends Bob $A = g^a \bmod p$
 - $A = 5^4 \bmod 23 = 4$ (in this example both A and a have the same value 4, but this is usually not the case)
3. Bob chooses a secret integer $b = 3$, then sends Alice $B = g^b \bmod p$
 - $B = 5^3 \bmod 23 = 10$
4. Alice computes $s = B^a \bmod p$
 - $s = 10^4 \bmod 23 = 18$
5. Bob computes $s = A^b \bmod p$
 - $s = 4^3 \bmod 23 = 18$
6. Alice and Bob now share a secret (the number 18).

Public Key Encryption

- Before you do asymmetric encryption, you generate a pair of keys:
 - Public-key
 - Private-key
- You broadcast your *public key* to everyone who wants to send you a message, so your friend can encrypt the message using your public key.
 - **Certificate Authorities** may be needed to ensure that the public key you proclaim to be yours, is actually yours.
 - They issue a **certificate** which contains owner's name and his/her public-key to anyone who wants to verify the correctness of it.
- Once you receive the encrypted message, you can decrypt it using your *private key* which is available only to you, and no one else.

Public Key Encryption



Digital Signatures

- We can reverse the public-key encryption to ensure authenticity of messages.
- We can use private-key to encrypt the message we want to send.
 - The encrypted message is called ***digital signature***.
- Everyone else can use our public-key to decrypt the message.
- Because we broadcast our public-key to everyone, and there are certification authorities who can prove that particular public-key belongs to us, and only to us, we can be sure that this message is originated from you, not anyone else.
- This plays the role of a traditional signature at the end of each letter, which can be associated to some person.

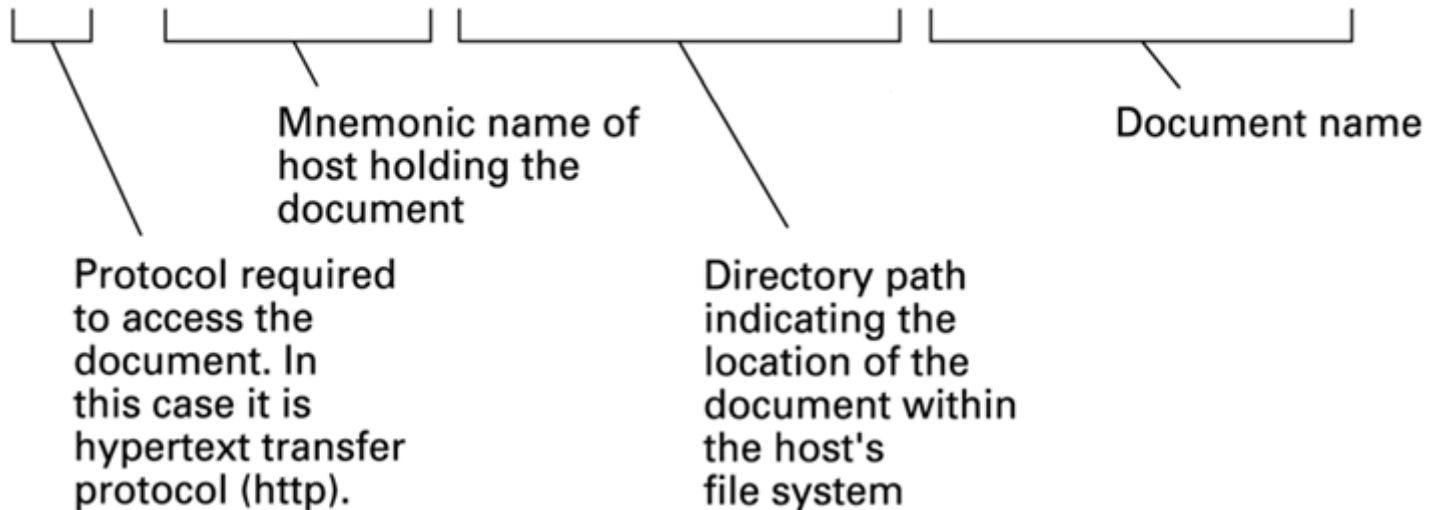
Basic HTML and CSS

World Wide Web

- **Hypertext** combines internet technology with concept of linked-documents
 - Embeds **hyperlinks** to other documents
- **Browsers** present materials to the user, installed on user's computer.
- **Webservers** provide access to documents, installed on a remote server.
- Documents are identified by **URLs** and transferred using **HTTP**

A typical URL

`http://eagle.mu.edu/authors/Shakespeare/Julius_Cesar.html`

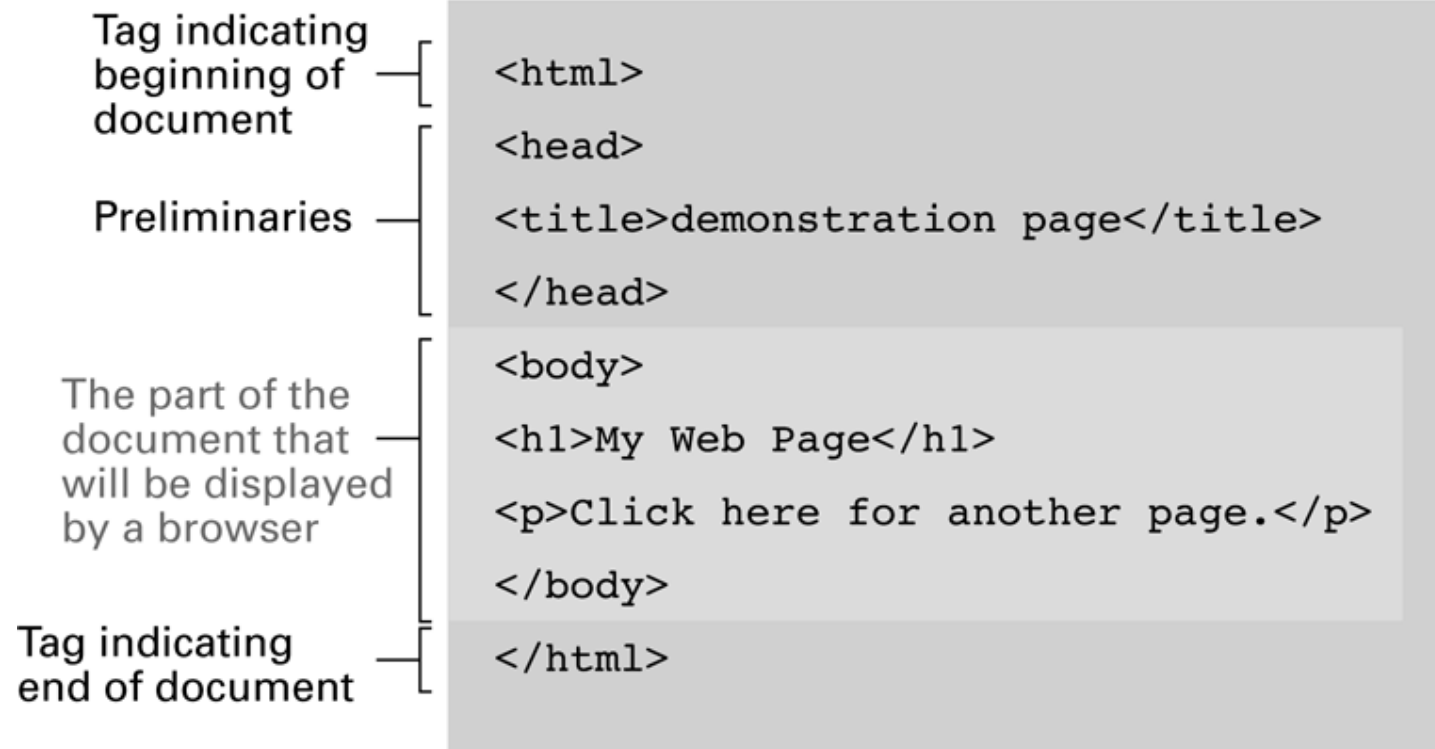


Hypertext Markup Language (HTML)

- Encoded as text file
- Contains tags to communicate with browser
 - Appearance
 - `<h1>` to start a level one heading
 - `<p>` to start a new paragraph
 - Links to other documents and content
 - `<a href = . . . >`
 - Insert images
 - `<img src = . . . >`
- We can learn basic HTML from W3C's tutorial web site or from many more [other web resources](#).

A simple webpage

a. The page encoded using HTML.



A simple webpage (continued)

b. The page as it would appear on a computer screen.



An enhanced simple webpage

a. The page encoded using HTML.

Anchor tag
containing
parameter — [

Closing
anchor tag — [

```
<html>
<head>
<title>demonstration page</title>
</head>
<body>
<h1>My Web Page</h1>
<p>Click
    <a href="http://crafty.com/demo.html">
    here
    </a>
    for another page.</p>
</body>
</html>
```

An enhanced simple Web page (continued)

b. The page as it would appear on a computer screen.



More HTML tags

- Some tags can contain additional information called attributes
 - syntax **<element attribute="value" attribute="value"> content </element>**
 - example: **Next page**

More HTML tags

- Some tags don't contain content; can be opened and closed in one tag

- syntax:

- `<element attribute="value" attribute="value" />`**

- example: **`<hr />`**

- example:

- ``**

Links <a>

```
<p>  
Search  
<a href="http://www.google.com/">Google</a>  
now!  
</p>
```

HTML

Search [Google](http://www.google.com/) now!

output

- The **href** attribute specifies the destination URL
- Links or *anchors* are inline elements, so they must be placed inside a block element such as a p or h1

More about anchors

```
<p><a href="deathlyHallows-book.html">Harry Potter and the  
Deathly Hallows Book</a></p>
```

```
<p><a href="http://en.wikipedia.org"  
title="Search">Wikipedia</a></p>
```

HTML

Harry Potter and the Deathly Hallows

Wikipedia_

output

- Types of URLs that can appear in anchors:
 - Absolute: to another web site
 - Relative: to another page on this web site

Nesting tags

Bad

```
<p>  
<a href=" deathlyHallows-book.html"> Harry Potter and the  
Deathly Hallows Book </p>  
<p>  
This text also links to Harry Potter Book</a>  
</p>
```

HTML

- Tags must be correctly nested: a closing tag must match the **most recently opened tag**
- The browser may render it correctly anyway, but it is invalid XHTML

Images

```

```

HTML



- The src attribute specifies source of the image URL
- XHTML also requires an alt attribute describing the image

More about images

```
<a href="http://harrypotter.net/">  
  
</a>
```

HTML



- If placed inside an a anchor, the image will become a link
- The title attribute specifies an optional tooltip

Line Break


```
<p>One Ring to rule them all, One Ring to find them, <br  
> One Ring to bring them all and in the darkness bind  
them.</p>
```

```
<p> In the Land of Mordor where the Shadows lie. </p>
```

HTML

One Ring to rule them all, One Ring to find them,
One Ring to bring them all and in the darkness bind them

In the Land of Mordor where the Shadows lie.

output

- br should be immediately closed with />
- br should not be used to separate paragraphs or used multiple times in a row to create spacing

Unordered list: ,

```
<ul>  
<li>No shoes</li>  
<li>No shirt</li>  
<li>No problem!</li>  
</ul>
```

HTML

- No shoes
- No shirt
- No problem!

output

- **ul** represents a bulleted list of items (block)
- **li** represents a single item within the list (block)

More about unordered lists

```
<ul>
<li>Harry Potter characters:
<ul>
<li>Harry Potter</li>
<li>Hermione</li>
<li>Ron</li>
</ul>
</li>
<li>LOTR characters:
<ul>
<li>Frodo</li>
<li>Bilbo</li>
<li>Sam</li>
</ul>
</li>
</ul>
```

HTML

More about unordered lists (cont.)

- Harry Potter characters:
 - Harry Potter
 - Hermione
 - Ron
- LOTR characters:
 - Frodo
 - Bilbo
 - Sam

output

Ordered list

```
<p>Apple business model:</p>
<ol>
<li>Beat Microsoft</li>
<li>Beat Google</li>
<li>Conquer the world!</li>
</ol>
```

HTML

Apple business model:

1. Beat Microsoft
2. Beat Google
3. Conquer the world

output

- **ol** represents a numbered list of items
- we can make lists with letters or Roman numerals using CSS (later)

Common error: Not closing a list

```
<ul>  
<li>No shoes</li>  
<li>No shirt</li>  
<li>No problem!</li>  
<p>Paragraph after list...</p>
```

HTML

- No shoes
- No shirt
- No problem!

Paragraph after list...

output

- If you leave a list open, subsequent contents will be indented

Common Error: Improper nested list placement

```
<ul>
<li>Harry Potter characters:</li>
<ul>
<li>Harry Potter</li>
<li>Hermione</li>
<li>Ron</li>
</ul>
</li>
<li>LOTR characters:
<ul>
<li>Frodo</li>
<li>Bilbo</li>
<li>Sam</li>
</ul>
</ul>
```

HTML

- closing the outer li too early (or not at all) will render correctly in most browsers, but it is incorrect XHTML

Definition list <dl>, <dt>, <dd>

```
<dl>
<dt>newbie</dt> <dd>one who does not have mad skills</dd>
<dt>jaded</dt> <dd>tired, bored, or lacking enthusiasm
</dd>
<dt>frag</dt> <dd>a kill in a shooting game</dd>
</dl>
```

HTML

newbie	one who does not have mad skills
jaded	Tired, bored, or lacking enthusiasm
frag	a kill in a shooting game

output

- **dl** represents a list of definitions of terms
- **dt** represents each term, and **dd** its definition

Tables <table>, <tr>, <td>

```
<table>
  <tr><td>1,1</td><td>1,2 okay</td></tr>
  <tr><td>2,1 real wide</td><td>2,2</td></tr>
</table>
```

HTML

1,1	1,2 okay
2,1 real wide	2,2

output

- ❑ `table` defines the overall table, `tr` each row, and `td` each cell's data
- ❑ Useful for displaying large row/column data sets
- ❑ NOTE: tables are sometimes used by novices for web page layout, but this is not proper semantic HTML and should be avoided

Table headers, captions: <th>, <caption>

```
<table>
  <caption>My important data</caption>
  <tr><th>Column 1</th><th>Column 2</th></tr>
  <tr><td>1,1</td><td>1,2 okay</td></tr>
  <tr><td>2,1 real wide</td><td>2,2</td></tr>
</table>
```

HTML

My important data

Column 1

Column 2

1,1

1,2 okay

2,1 real wide

2,2

output

- ❑ th cells in a row are considered headers
- ❑ a caption at the start of the table labels its meaning

Cascading Style Sheets (CSS)

- Describes the appearance, layout, and presentation of information on a web page
 - HTML describes **the content** of the page
- Describes *how* information is to be displayed, not *what* is being displayed
- Can be embedded in HTML document or placed into separate .css file

Basic CSS rule syntax

```
selector {  
property: value;  
property: value;  
...  
property: value;  
}
```

CSS

```
p {  
font-family: sans-serif;  
color: red;  
}
```

CSS

- A CSS file consists of one or more **rules**
- Each rule starts with a **selector**
- A selector specifies an HTML element(s) and then applies style **properties** to them
 - a selector of `*` selects all elements

Attaching a CSS file <link>

```
<head>
...
<link href="filename" type="text/css" rel="stylesheet" />
...
</head>
```

HTML

```
<link href="style.css" type="text/css" rel="stylesheet" />
<link href="http://www.google.com/uds/css/gsearch.css"
rel="stylesheet" type="text/css" />
```

HTML

- A page can link to multiple style sheet files
 - In case of a conflict (two sheets define a style for the same HTML element), the latter sheet's properties will be used

Embedding style sheets: <style>

```
<head>
<style type="text/css">
p { font-family: sans-serif; color: red; }
h2 { background-color: yellow; }
</style>
</head>
```

HTML

- CSS code can be embedded within the head of an HTML page
- *Bad style* and should be avoided when possible (why?)

Inline styles: the style attribute

```
<p style="font-family: sans-serif; color: red;">  
This is a paragraph</p>
```

HTML

This is a paragraph

output

- Higher precedence than embedded or linked styles
- Used for one-time overrides and styling a particular element
- *Bad style* and should be avoided when possible (why?)

CSS properties for colors

```
p {  
  color: red;  
  background-color: yellow;  
}
```

CSS

This paragraph uses the style above

output

property	description
color	color of the element's text
background-color	color that will appear behind the element

Specifying colors

```
p { color: red; }  
h2 { color: rgb(128, 0, 196); }  
h4 { color: #FF8800; }
```

CSS

This paragraph uses the first style above

This h2 uses the second style above.

This h4 uses the third style above.

output

- **color names:** aqua, black, blue, fuchsia, gray, green, lime, maroon, navy, olive, purple, red, silver, teal, white (white), yellow
- **RGB codes:** red, green, and blue values from 0 (none) to 255 (full)
- **hex codes:** RGB values in base-16 from 00 (0, none) to FF (255, full)

Grouping styles

```
p, h1, h2 {  
  color: green;  
}  
h2 {  
  background-color: yellow;  
}
```

CSS

This paragraph uses the above style.

This h2 uses the above styles.

output

- A style can select multiple elements separated by commas
- The individual elements can also have their own styles

CSS class selectors

```
<p class="shout">Coding Horror! Coding Horror!</p>  
<p class="special">See our special deal on Droids!</p>  
  
<p class="special">Today only!</p>
```

HTML

```
.special {  
background-color: yellow;  
font-weight: bold;  
}  
  
p.shout {  
color: red;  
font-family: cursive;  
}
```

CSS

Coding Horror! Coding Horror!

See our special deal on Droids!

Today only!

output

CSS properties for fonts

property	description
font-family	which font will be used
font-size	how large the letters will be drawn
font-style	used to enable/disable italic style
font-weight	used to enable/disable bold style

[Complete list of all CSS properties](http://www.w3schools.com/css/css_reference.asp) (http://www.w3schools.com/css/css_reference.asp)

Client Side vs Server Side

- To create multimedia-rich interactive web pages, you may require *client-side* and *server-side activities*.
- Client-side activities
 - **JavaScript** – AngularJS, ReactJS, BackboneJS, KnockoutJS, etc
 - **HTML5, CSS3** – Bootstrap, Ionic, Material Design, Polymer, etc
 - **Rich-clients** - Silverlight, Adobe Flash, Java Applets
- Server-side activities
 - Common Gateway Interface (CGI)
 - **Server Pages:** PHP, NodeJS, ASP.NET MVC, JavaServer Pages (JSP), Active Server Pages (ASP), etc
 - **Web Services:** Web API, NodeJS, Flask (Python), JAX-WS/JAX-RS, etc

Suggested Reading/Activity

- Learn HTML and CSS:
 - www.w3schools.com/html
 - eDX online courses for web development:
<https://www.edx.org/school/w3cx>
- Capturing Internet Packets:
 - Install Wireshark: <http://www.wireshark.org/>
 - Follow [this tutorial](#) on using Wireshark
- Watch over the weekend:
 - [What is Internet?](#) – a 7 video episodes from the creators of Internet.